

Project Initialization and Planning Phase

Date	24 June 2026
Team ID	XXXXXXXXXXXX
Project Title	Credit Card Approval Prediction
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed project aims to automate the credit card approval process using machine learning techniques, improving speed and accuracy in decision-making. The system reduces manual work, minimizes human errors, and helps banks identify eligible and high-risk applicants efficiently. Key features include machine learning-based prediction and real-time approval decisions through a web application.

Project Overview	
Objective	The primary objective is to develop an intelligent system that predicts whether a credit card application will be approved or rejected using machine learning algorithms, ensuring faster and more accurate decisions.
Scope	The project analyzes applicant financial and demographic data and provides an automated credit card approval system through a Flask web application and cloud deployment.
Problem Statement	
Description	The existing credit card approval process is time-consuming and prone to human errors because applications are manually reviewed.
Impact	Solving these issues will improve operational efficiency, reduce processing time, minimize risks, and enhance customer satisfaction.
Proposed Solution	
Approach	Use machine learning algorithms such as Logistic Regression, Random Forest, Decision Tree, and XGBoost to analyze applicant data and predict credit card approval or rejection.
Key Features	- Automated credit card approval prediction. - Real-time decision-making through a web application. - Identification of high-risk applicants.

	• User-friendly interface for bank staff and customers. • Cloud deployment using IBM Watson Machine Learning.
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Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Hardware	Computing Resources	CPU/GPU (Google Colab T4 GPU or equivalent)
Hardware	Memory	8 GB RAM
Hardware	Storage	1 TB SSD
Software		
Software	Framework	Flask
Software	Libraries	scikit-learn, pandas, numpy, matplotlib, xgboost
Software	Development Environment	Jupyter Notebook, VS Code/PyCharm
Data		
Data	Dataset	Credit Card Approval Dataset (CSV format) containing applicant financial and demographic information